Differentiation — Product Rule Skill Ladder

NCEA Level 3 Maths, AS91578 · one tiny skill at a time · do not skip ahead

Name: _____ Date: _____

How to use this sheet: Do Chunk 0 completely (read the explanation, do all 10 questions, check them against the answer sheet) before looking at Chunk 1. If you get more than 2 wrong in a chunk, stop and redo that chunk's explanation before moving on. This sheet comes after the Chain Rule Skill Ladder — if $\sec^2 x$, or finding the derivative of a plain function like x^3 or $\sin x$, isn't instant for you yet, go back to that sheet first.

The ladder: Chunk 0 spot the two factors → Chunk 1 differentiate each factor alone → Chunk 2 plug given pieces into the formula → Chunk 3 full product rule, simple one-term factors → Chunk 4 full product rule, multi-term factors → Chunk 5 full product rule, mixed function types. Every factor on this sheet is deliberately kept plain (no chain rule needed inside a factor) — combining chain rule AND product rule together is a later sheet, once both are separately solid.

Chunk 0 — Spot the two factors being multiplied

The rule: a product-rule question always has TWO separate functions of x multiplied together. Name the first one (as written, left to right) "u" and the second one "v". Staying consistent with left=u, right=v stops you mixing yourself up later — the final answer comes out the same either way, but consistency helps while you're still learning this.

WORKED EXAMPLE

For $y = x^3 \sin x$, name u and v.

Answer: $u = x^3$, $v = \sin x$

Your turn — name u (first factor) and v (second factor). Don't differentiate anything yet.

1. $x^2 e^x$

u = v =

2. $x \cdot \ln x$

u = v =

3. $\sin x \cdot \cos x$

u = v =

4. $x^4 \tan x$

u = v =

5. $e^x \cos x$

u = v =

6. $(x+1)(x-3)$

u = v =

7. $x \cdot e^x$

u = v =

8. $\ln x \cdot \sin x$

u = v =

9. $x^2(x^3-2)$

u = v =

10. $\cos x \cdot x^5$

u = v =

Chunk 1 — Differentiate each factor on its own

The rule: before you can assemble the product rule, you need u' and v' ready to go. This is plain, non-composite differentiation — the same basic facts from the Chain Rule Skill Ladder's Chunk 1. Get these instant, because you'll need a pair of them every single time.

WORKED EXAMPLE

For $u = x^3$, find u'.

Answer: $u' = 3x^2$

Your turn — differentiate each factor.

1. $u = x^2$

u' =

2. $v = e^x$

v' =

3. $u = x$

u' =

4. $v = \ln x$

v' =

5. $u = \sin x$

u' =

6. $v = \cos x$

v' =

7. $u = x^4$

u' =

8. $v = \tan x$

v' =

9. $u = x+1$

u' =

10. $v = x-3$

v' =

Chunk 2 — Plug given pieces into the product rule formula

The product rule: if y = uv, then dy/dx = u'v + uv' — "derivative of the first times the second, PLUS the first times derivative of the second." In this chunk, u, v, u' and v' are all given to you already — your only job is to plug them into the formula correctly and simplify. Get this pattern automatic before you ever have to differentiate anything yourself.

WORKED EXAMPLE

$u=x^2$, $v=\sin x$, $u'=2x$, $v'=\cos x$. Find dy/dx for $y=uv$.

$dy/dx = u'v + uv' = (2x)(\sin x) + (x^2)(\cos x)$

Answer: $dy/dx = 2x\cdot\sin x + x^2\cos x$

Your turn — plug the given pieces into u'v + uv' and simplify.

1. $u=x^3$, $v=e^x$, $u'=3x^2$, $v'=e^x$

2. $u=x$, $v=\ln x$, $u'=1$, $v'=1/x$

3. $u=\sin x$, $v=\cos x$, $u'=\cos x$, $v'=-\sin x$

4. $u=x^4$, $v=\tan x$, $u'=4x^3$, $v'=\sec^2 x$

5. $u=e^x$, $v=\cos x$, $u'=e^x$, $v'=-\sin x$

6. $u=x+1$, $v=x-3$, $u'=1$, $v'=1$

7. $u=x$, $v=e^x$, $u'=1$, $v'=e^x$

8. $u=\ln x$, $v=\sin x$, $u'=1/x$, $v'=\cos x$

9. $u=x^2$, $v=x^3-2$, $u'=2x$, $v'=3x^2$

10. $u=\cos x$, $v=x^5$, $u'=-\sin x$, $v'=5x^4$

Chunk 3 — Full product rule, simple one-term factors

The 4-step method, now from scratch:
1. Name u and v (Chunk 0).
2. Find u' and v' (Chunk 1).
3. Plug into dy/dx = u'v + uv' (Chunk 2).
4. Tidy up.

WORKED EXAMPLE

Differentiate $y = x^2e^x$.

Step 1: $u=x^2$, $v=e^x$.

Step 2: $u'=2x$, $v'=e^x$.

Step 3: $dy/dx = 2x\cdot e^x + x^2\cdot e^x$.

Answer: $dy/dx = x^2e^x + 2xe^x$

Your turn — full computation, 4-step method, each factor is a single term.

1. $y = x^3\cdot\sin x$

2. $y = x\cdot e^x$

3. $y = x^2\cdot\ln x$

4. $y = e^x\cdot\cos x$

5. $y = x^4\cdot\cos x$

6. $y = \sin x\cdot\ln x$

7. $y = x \cdot \tan x$

8. $y = e^{x} \cdot \sin x$

9. $y = x^{2} \cdot \cos x$

10. $y = \ln x \cdot e^{x}$

Chunk 4 — Full product rule, multi-term factors

Same 4-step method — but now each factor has more than one term, so differentiate the WHOLE factor (term by term) before assembling. For two pure polynomials, you can double-check your answer by expanding the brackets first and differentiating that directly — both methods must give the same answer.

WORKED EXAMPLE

Differentiate $y = (x^2+1)(x-3)$.

Step 1–2: $u=x^2+1$, $u'=2x$. $v=x-3$, $v'=1$.

Step 3: $dy/dx = 2x(x-3) + (x^2+1)(1) = 2x^2-6x+x^2+1$

Check by expanding: $y=x^3-3x^2+x-3 \rightarrow dy/dx=3x^2-6x+1$ ✓ matches.

Answer: $dy/dx = 3x^2-6x+1$

Your turn — full computation. Try expanding-and-checking on a couple of these if you're unsure.

1. $y=(x+1)(x^2-3)$

2. $y=(2x-5)(x^2+x)$

3. $y=(x^2+2)(3x-1)$

4. $y=(x^3-1)(x+4)$

5. $y=(x^2-x)(x+2)$

6. $y=(5-x)(x^2+3)$

7. $y=(x^2+4x)(x-2)$

8. $y=(3x+2)(x^2-5)$

9. $y=(x-7)(x^2+x+1)$

10. $y=(x^2-9)(2x+1)$

Chunk 5 — Full product rule, mixed function types

Same 4-step method, one more time — mixing different "flavours" of function together (trig×exponential, ln×trig, trig×trig, and so on) so the pattern becomes automatic no matter which two kinds of function you're handed.

WORKED EXAMPLE

Differentiate $y = \tan x \cdot \ln x$.

$u=\tan x$, $u'=\sec^2 x$. $v=\ln x$, $v'=1/x$.

Answer: $dy/dx = \sec^2 x \cdot \ln x + \tan x \cdot (1/x)$

Your turn — full computation.

1. $y = \cos x \cdot e^x$

2. $y = x^3 \cdot \tan x$

3. $y = \ln x \cdot \cos x$

4. $y = \sin x \cdot e^x$

5. $y = x^5 \cdot \ln x$

6. $y = \tan x \cdot \cos x$

7. $y = e^{x} \cdot \ln x$

8. $y = \sin x \cdot x^{2}$

9. $y = \cos x \cdot \tan x$

10. $y = \ln x \cdot x^{4}$